

CCE-423 Wireless Communication Midterm Syllabus.

1. Explain the key differences between wireless communications and cellular communications, highlighting how each technology is utilized in modern communication systems.

Aspect	Wireless Communication	Cellular Communication
Meaning	Any communication without wires (Wi-Fi, Bluetooth, Satellite).	A type of wireless made for mobile phones using cell towers.
Range	Can be short (Wi-Fi, Bluetooth) or very wide (satellite).	Wide coverage divided into cells with towers.
Mobility	Limited mobility (e.g., Wi-Fi doesn't work everywhere).	Full mobility – works while moving from place to place.
Setup	Simple (router, hotspot, etc.).	Complex (towers, base stations, network).
Use	Internet at home, sharing files, satellite links.	Mobile calls, SMS, 4G/5G internet.
Examples	Wi-Fi, Bluetooth, Satellite.	GSM, LTE, 5G.

2.

- i. Are we currently operating at the Shannon limit in the physical layer of communication systems? What would Shannon's perspective be on this?

- **Current status:** No, we are not exactly operating at the Shannon limit in real systems. Modern coding techniques (like Turbo codes and LDPC codes) get close, but practical issues such as noise, hardware limitations, and complexity prevent reaching the theoretical maximum.
- **Shannon's perspective:** He would likely be impressed at how close current systems (4G, 5G, Wi-Fi 6/7) operate to his limit, but also note that there is still a gap between theory and practice due to real-world constraints.

- ii. Additionally, what factors contribute to the reduction in Wi-Fi performance?" How can we overcome this limitation?

- **Factors reducing Wi-Fi performance:**

1. Interference from other devices (microwaves, Bluetooth, other Wi-Fi networks).
2. Physical obstacles (walls, furniture).
3. Limited frequency spectrum and congestion in busy areas.
4. Signal attenuation with distance.
5. Old hardware or outdated Wi-Fi standards.

- **Ways to overcome limitations:**

1. Use dual-band or tri-band routers (2.4 GHz, 5 GHz, 6 GHz).
2. Place routers in central, open spaces.
3. Use mesh Wi-Fi systems for larger coverage.
4. Upgrade to latest standards (Wi-Fi 6/6E).
5. Reduce interference by changing Wi-Fi channel or frequency band.

3. Write down the objectives and learning outcomes of wireless and cellular communication course?

Objectives

1. To introduce the fundamental concepts of wireless communication systems.
2. To understand the principles and architecture of cellular networks.
3. To study multiple access techniques (FDMA, TDMA, CDMA, OFDMA).
4. To explore different generations of cellular networks (1G–5G).
5. To understand design issues like handoff, capacity, and coverage.
6. To prepare students for practical applications in modern wireless systems (Wi-Fi, LTE, 5G, IoT).

Learning Outcomes: After completing this course, students will be able to:

1. Explain the basics of wireless and cellular communication systems.
2. Analyze radio channel characteristics such as path loss, fading, and interference.
3. Compare different multiple access and modulation techniques.
4. Understand cellular concepts like frequency reuse, handoff, and capacity planning.
5. Describe the evolution of cellular technologies (from 1G to 5G).
6. Evaluate the performance of wireless networks and suggest improvements.
7. Apply knowledge of wireless systems in real-world applications (Wi-Fi, mobile networks, IoT).

4. What do you mean by wireless communication and cellular communication.

Wireless communications is a type of data communication that is performed and delivered wirelessly. This is a broad term that incorporates all procedures and forms of connecting and communicating between two or more devices using a wireless signal through wireless communication technologies and devices.

A cellular network or mobile network is a communication network where the last link is wireless. The network is distributed over land areas called cells, each served by at least one fixed-location transceiver, known as a cell site or base station.

5. Explain the strategies of different network operators while migrating toward 3G to 4G communication system. Why is a single communication system not in real world?

Or

Check out the strategies of different network operators while migrating towards third generation systems. Why is a single common system not in sight?

Migration Strategies (3G → 4G)

1. **Overlay** – Keep 3G, add 4G as a new layer.
2. **Refarming/Upgrade** – Reuse old 3G spectrum for 4G.
3. **Parallel Networks** – Run 3G & 4G together (voice on 3G, data on 4G).
4. **Greenfield** – New operators directly launch 4G (skip 3G).

Why no single system in real world?

- Old phones still need 2G/3G (backward support).
- 2G/3G cover rural areas cheaply.
- Each system has unique roles (2G = voice, 3G = moderate data, 4G = high-speed data).
- Not all users upgrade at the same time.

6. Make a comparison table of different wireless system generation.

Comparison table of different wireless communication system generations (1G to 6G):

Gen	Period	Tech	Speed	Highlights	Limits
1G	1980s	Analog (FDMA)	~2.4 Kbps	Voice only, analog signals, first mobile phones	Poor quality, no security
2G	1990s	GSM, CDMA, TDMA	≤64 Kbps	Digital calls, SMS/MMS, better security	Low data, limited media
2.5G	Late 1990s	GPRS, EDGE	56–384 Kbps	Basic internet, email, WAP services	Slow for multimedia
3G	2000s	WCDMA, CDMA2000, HSPA	384 Kbps–2 Mbps	Internet, video calls, mobile TV	High battery use, limited BW
4G	2010s	LTE, WiMAX (OFDMA, MIMO)	100 Mbps–1 Gbps	HD streaming, VoIP, gaming, better efficiency	Costly infra, rural issues
5G	2020s	Massive MIMO, mmWave	1–20 Gbps	Ultra-fast, low latency, IoT, AR/VR, smart cars	Limited coverage, expensive
6G	~2030s	Terahertz, AI-driven	Up to 1 Tbps	AI networks, holograms, brain–computer link	Still theoretical

7. Distinguished between licensed and unlicensed band with example. Why does Wi-Fi performance reduce?

Feature	Licensed Band	Unlicensed Band
Definition	Frequency spectrum that requires government permission (license) to operate.	Frequency spectrum that anyone can use without a license, following certain rules.
Cost	Expensive; operators pay fees for using the spectrum.	Free to use, but must follow regulations (e.g., power limits).
Interference	Minimal, as operators have exclusive rights.	High potential, since multiple devices share the band.
Control	Regulated and monitored by authorities.	Shared among multiple users; no exclusive control.
Example	900 MHz GSM, 1800 MHz LTE	2.4 GHz and 5 GHz Wi-Fi, Bluetooth

Factors reducing Wi-Fi performance:

Same as 2(ii)

1. Interference from other devices (microwaves, Bluetooth, other Wi-Fi networks).
 2. Physical obstacles (walls, furniture).
 3. Limited frequency spectrum and congestion in busy areas.
 4. Signal attenuation with distance.
 5. Old hardware or outdated Wi-Fi standards.
8. **Discover the current numbers of subscribers for the different systems. As mobile communications boom, no printed number is valid for too long!**

Current Mobile Subscriber Numbers (2025)

- **Worldwide**
 - **Total mobile users:** About **8.8 billion**
 - **5G users:** ~**2.4 billion** (growing very fast)
 - **4G users:** ~**4.9 billion** (slowly decreasing)
 - **3G users:** ~**0.7 billion** (shrinking)
 - **2G users:** Only small numbers left, many countries are shutting it down.

Why “no printed number stays valid for long”.**Mobile communication is changing extremely fast:**

- Massive quarterly shifts—like **145 million new 5G users in just Q1 2025**—quickly alter the landscape [TelecomLead](#).
- Operators are **shutting down older networks** (3G and 2G), accelerating migration to newer technologies.
- Reports from sources like **Ericsson, GSMA, and Omdia** are essential for **up-to-the-minute data**.

❖ Key Technologies of wireless communication and components.**Key Technologies of wireless communications**

1. Radio frequency (RF)
2. Infrared (IR)
3. Microwave

Components:

1. Transmitter: convert data→
Electromagnetic Sig.
2. Receiver: decode transmitted signals
3. Antenna Modulation Technique.

❖ Challenges and advancements of wireless communications:**Challenges of wireless communication**

1. Interference
2. Signal attenuation
3. Security
4. Spectrum allocation.

Advancement:

1. Faster data rate
2. Imense bandwidth
3. Reliability enhancement
4. Enhanced security.

❖ Spectrum allocation of Bangladesh mobile operators.

Spectrum Allocation (Aug 2024)

- Grameenphone (GP) → 127.4 MHz → ~17 lakh users per MHz
- Robi Axiata → 124 MHz → ~12 lakh users per MHz
- Banglalink → 80 MHz → ~9 lakh users per MHz
- Teletalk → 55.2 MHz → ~2.6 lakh users per MHz

❖ Challenges of wireless network/radio and SoC/Device.

Challenges of wireless network/radio

1. Gbps data rates with no error
2. Energy efficiency
3. Scarce/fragmented spectrum
4. Reliability and convergence
5. Heterogeneous network
6. Seamless internet hand off

SoC/Device Challenges

1. Performance
2. Complexity
3. Size, power, cost
4. High frequencies
5. Multiple antennas
6. Multiradio integration & coexistence

❖ What are pros and cons regarding push or pull services?

Type	Pros	Cons
Push (info sent automatically to user)	- Instant updates (e.g., notifications, emails, news alerts) - User doesn't need to check manually - Good for time-critical info	- Can overload users (too many notifications) - Wastes bandwidth if unwanted - Privacy concerns (tracking)
Pull (user requests info when needed)	- User in control - Less spam/noise - Efficient use of bandwidth	- Delayed info (not real-time) - User must take action to get updates - May miss urgent events

❖ 4G or higher coverage is flattening way before reaching the world population. Give reasons!

4G and higher network coverage is flattening before reaching the whole world mainly because of **economic, geographic, and social barriers**. Expanding coverage to remote and rural areas is very costly, while the return for operators is low due to fewer users and limited revenue. Difficult terrains like mountains, forests, or deserts make infrastructure deployment harder, and maintaining base stations without reliable power adds further challenges. In many regions, people also cannot afford modern smartphones or data plans, so companies see little incentive to invest. Political issues, spectrum regulations, and conflicts can also slow down rollout. In short, while cities enjoy fast networks, extending them everywhere is limited by cost, geography, and affordability.